

A very short guide to formatting HPSG with LaTeX

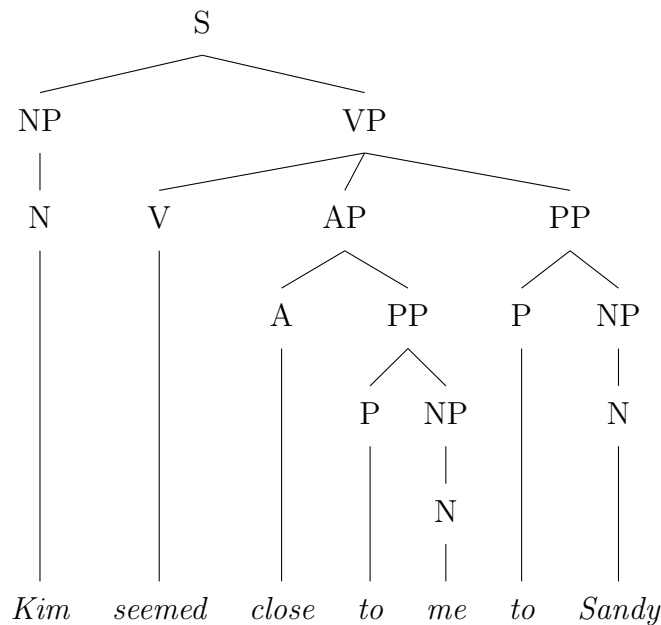
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Some recommended packages, and examples of trees and feature structures. For some advice on style take a look at my *(Computational) Linguistic Style Guidelines: a guide for the flummoxed*, or the very comprehensive *Generic Style Rules for Linguistics: Abridged LangSci version*.

1 Syntactic Tree

For trees, we use **forest**:



You can draw the trees using brackets.

2 AVM

We use **langsci-avm** for feature structures (attribute value matrices).

Lexical Entry for *eat*

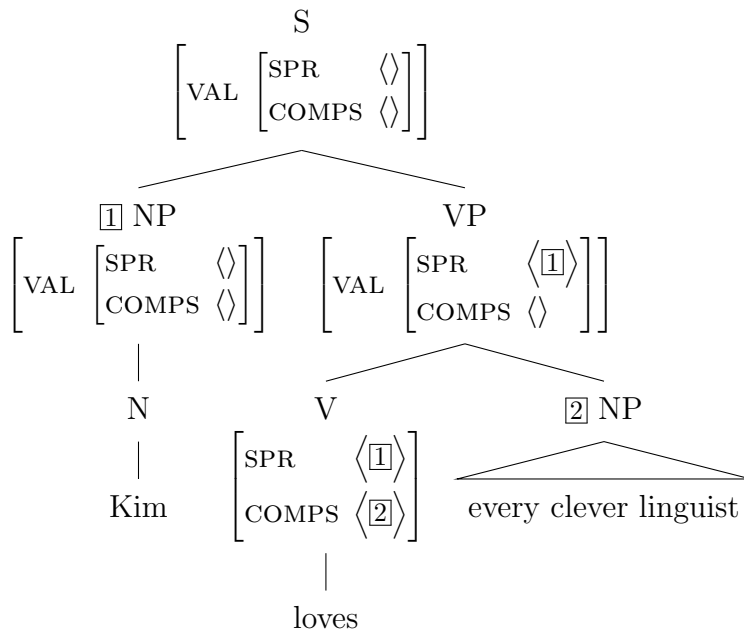
$$\left\langle \text{devour}, \begin{bmatrix} \text{word} \\ \text{HEAD} \text{ verb} \\ \text{VAL} \begin{bmatrix} \text{COMPS} \langle \text{NP} \rangle \end{bmatrix} \end{bmatrix} \right\rangle$$

Binary Head-Complement Rule

$$\begin{bmatrix} \text{phrase} \\ \text{VAL} \begin{bmatrix} \text{COMPS} \langle \rangle \end{bmatrix} \end{bmatrix} \rightarrow \mathbf{H} \begin{bmatrix} \text{word} \\ \text{VAL} \begin{bmatrix} \text{COMPS} \langle \boxed{1} \rangle \end{bmatrix} \end{bmatrix} \boxed{1}$$

3 Tree and AVM

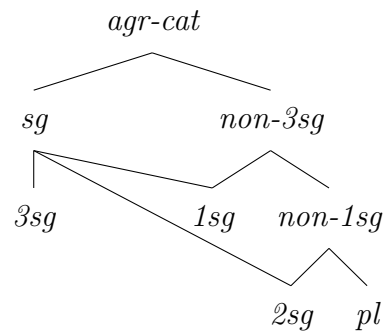
AVMs can be embedded in tree nodes:



It's easiest to build the tree first, then add the avms bit by bit, ...

4 Type Hierarchy

You can also add extra lines to trees, using the underlying package **tikz**. See the **forest** documentation for more details.



5 Numbered Examples and Glosses

I recommend using [langsci-gb4e](#) for numbered examples.

- (1) An example
- (2) Další příklad “Another example” (Czech)
- (3) Japanese
manmosu-wa zetumetu-shita
 mammoth-TOP die.out-did
 “Mammoths died out.”
- (4) Another example

You can refer to an example: (2) is an informal way, only ok if you know your readers will know the languages; (3) follows the Leipzig Glossing Rules.

In your assignment, you should gloss them with the [Leipzig Glossing Rules](#).